

Banking Analytics Dashboard using SQL & Power BI

Project Overview

- End to End Banking Analytics Project
- SQL + Power BI Workflow
- Customer & Transaction Analysis
- KPI Driven Dashboard

Business Problem

- Analyze Customer Behavior
- Monitor Transaction Trends
- Track Inactive Accounts
- Understand Account Performance
- Detect Financial Irregularities

Project Objectives

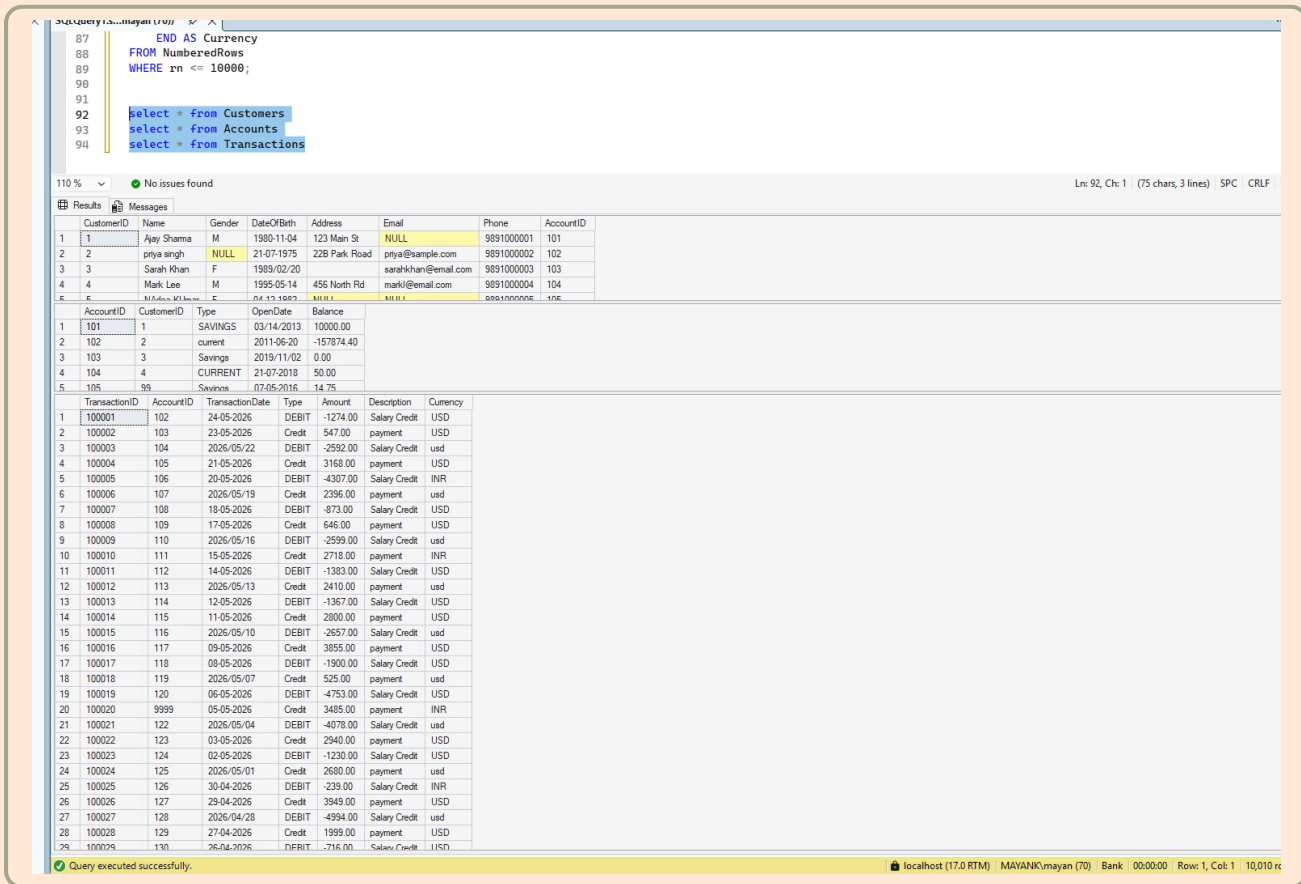
- Generate Synthetic Banking Data
- Simulate Real World Data Issues
- Clean & Standardize Data
- Build KPIs using DAX
- Create Interactive Dashboard
- Generate Business Insights

Tools & Technologies Used

- SQL Server
- Power BI Desktop
- Power Query
- DAX
- Notion
- XMind
- PowerPoint
- GitHub

Dataset Overview

- Customers Table
  - CustomerID
  - Name
  - Gender
  - DateOfBirth
  - Contact Details
- Accounts Table
  - AccountID
  - CustomerID
  - Account Type
  - OpenDate
  - Balance
- Transactions Table
  - TransactionID
  - AccountID
  - TransactionDate
  - Transaction Type
  - Amount
  - Currency



Data Generation

- Synthetic Data Creation
- AI Assisted Prompt Engineering
- Generated 10,000 Transactions
- Intentional Data Inconsistencies
- Realistic Banking Simulation

Data Quality Issues Introduced

- Mixed Date Formats
- Null Values
- Invalid Relationships
- Inconsistent Text Capitalization
- Duplicate Possibilities
- Outlier Balances
- Currency Inconsistencies

SQL Data Cleaning

- Standardized Date Formats
- TRY\_CONVERT()
- CASE Statements
- MM/dd/yyyy Formatting
- Validation Queries

SQL Joins & Data Integration

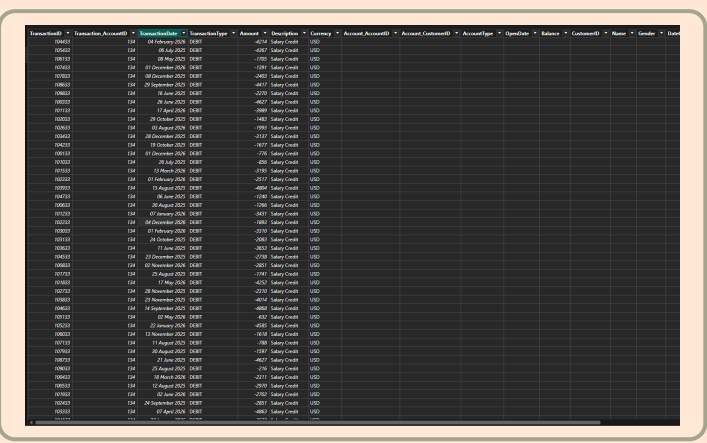
- LEFT JOIN Operations
- Transactions + Accounts
- Accounts + Customers
- CombinedBankingDataset
- Preserved Missing Relationships

Power BI Integration

- Connected SQL Server
- Imported Combined Dataset
- Power Query Editor
- Data Type Validation

Power Query Cleaning

- Replaced Null Descriptions
- Standardized Currency Values
- Fixed Account Type Cases
- Handled Null Account IDs
- Changed Data Types
- Locale Based Date Conversion

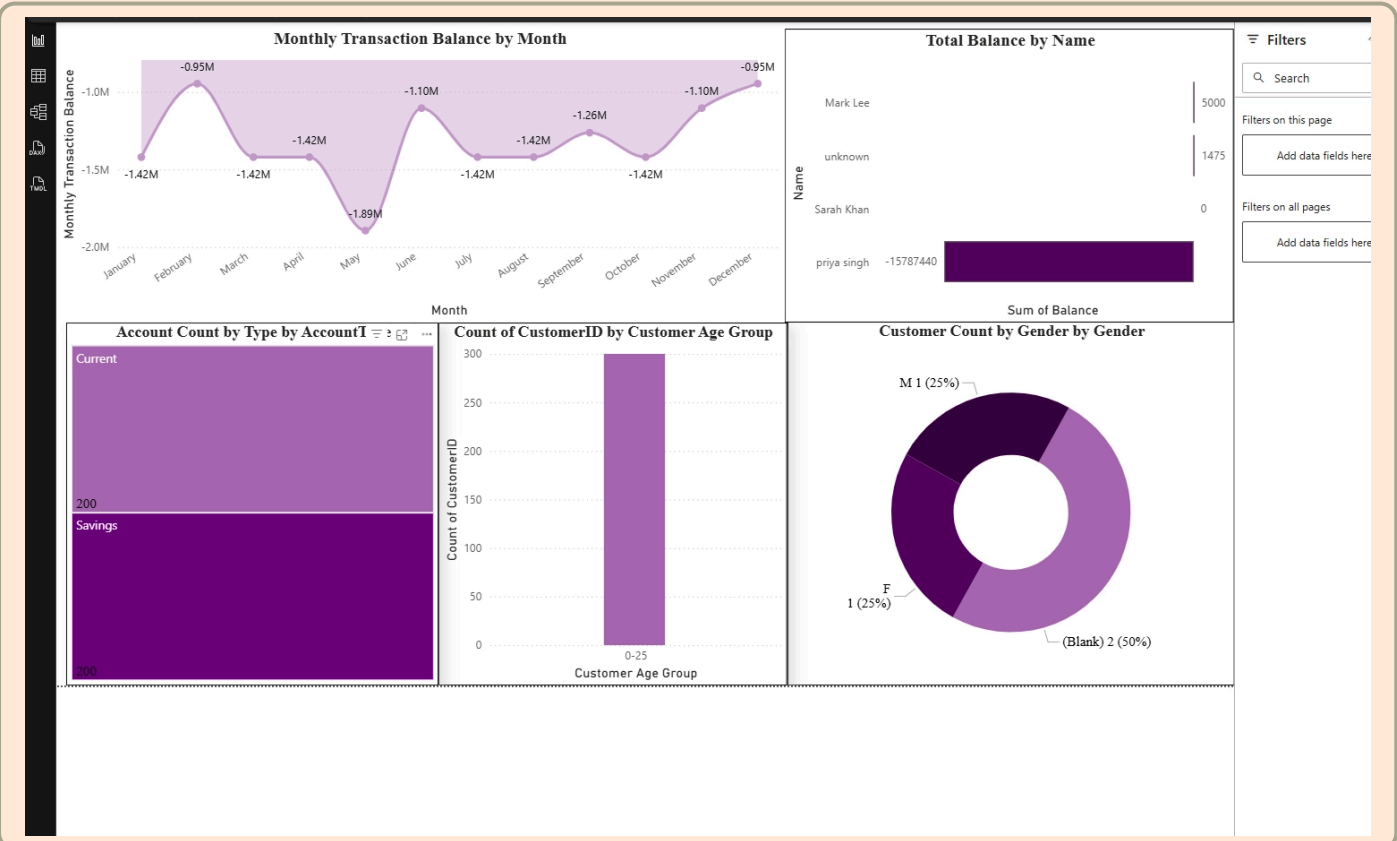
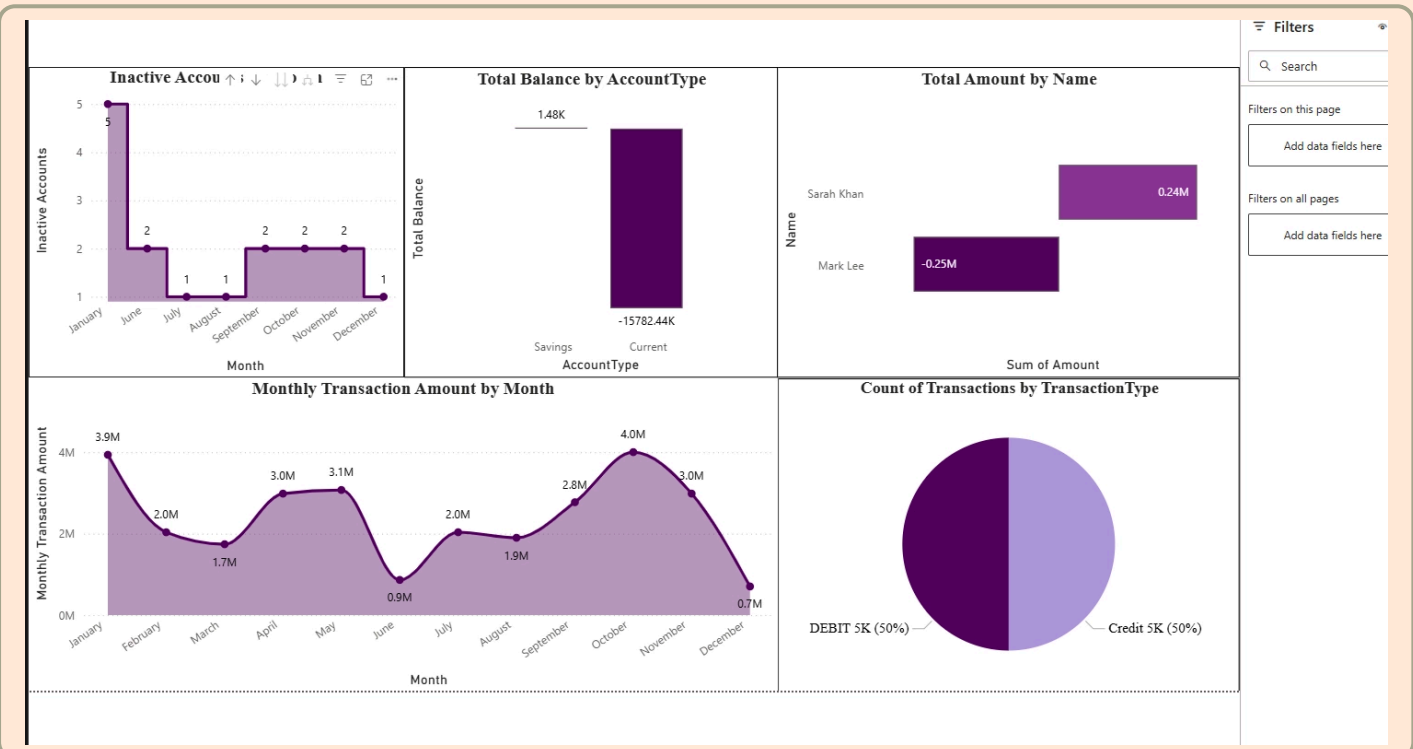


KPI Planning & Dashboard Design

- Customer KPIs
- Account KPIs
- Transaction KPIs
- Financial KPIs
- Business Oriented Visuals

Dashboard KPIs

- Transactions by Type
- Monthly Transaction Amount
- Total Balance by Account Type
- Total Balance by Name
- Inactive Accounts by Month
- Customer Gender Distribution
- Customers by Age Group
- Account Count by Type
- Transaction Volume Trend



Key Insights

- Equal Credit & Debit Distribution
- Monthly Transaction Fluctuations
- Inactive Account Detection
- Customer Demographic Analysis
- Account Type Distribution
- Outlier Detection